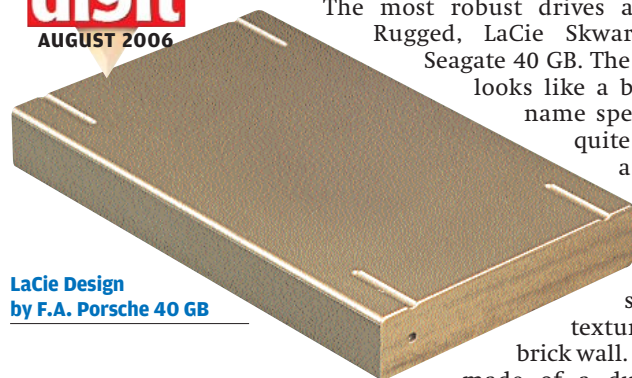




LaCie Design
by F.A. Porsche 40 GB

connectors to connect to two USB ports. This is done to remedy the same power problem mentioned, but in this case, there is a single cable that carries power as well as data.

The Transcend drives and the LaCie 40 GB each came with carry pouch.

Other Features

The most robust drives are the LaCie Rugged, LaCie Skwarim and the Seagate 40 GB. The LaCie Rugged looks like a brick, and the name spells it out: it's quite rugged! It has a rubber shock-absorbing jacket. LaCie Skwarim is a perfect square, and its texture looks like a brick wall. The exterior is made of a durable, shock-absorbent material. We had a hard time overcoming the urge to see if these drives could survive a "drop test"!

The Seagate has a hard casing with a unique design that looks like a wire mesh on the sides.

The aluminium casings of the Freecom are also quite sturdy, with a smooth finish. The WD Passport matched up to the Freecom, though it had a plastic casing.

The Transcend and Iomega drives, too, have aluminium cases, but these aren't as firm as those of the others. They can be easily dented—not a good thing, since these drives are meant to be carried around.

To be truly portable, a drive should, of course, be small and very light. The Freecom FHD-XS is the sleekest drive in this category, while the lightest is the LaCie Skwarim, weighing just 99 grams. The LaCie Rugged is the drive with the largest dimensions, but isn't the heaviest—that dubious honour goes to the Seagate 40 GB drive, at 292 grams. Of course, even at 292 grams, you can't consider this drive *heavy*!

The drives from LaCie are not only rugged, they are also elegantly designed. The LaCie Skwarim was designed by world-renowned product and interior designer Karim Rashid; it is pink, it's smooth to the touch, and it looks like it was designed specially for women.

The LaCie Design by F. A. Porsche hard drive was designed by the well-known design agency Porsche Design GmbH. These are drives you can actually show off!

How We Tested

Test Machine Configuration

Our test machine consisted of an Intel Pentium 4 processor running at 3.6 GHz on an Intel 925XCV motherboard with 1 GB of Micron DDR2 533 MHz RAM and a Seagate Barracuda 120 GB SATA hard drive. A fresh installation of Windows XP with SP2 was done and the latest versions of the hardware drivers were installed along with DirectX 9.0c.

Test Procedure

The performance tests consisted of synthetic tests using benchmarking software and real-world tests where application performance and file transfer speeds were tested.

The synthetic tests were conducted using HD Tach RW 3.0.1.0 and SiSoft Sandra 2007 Home, which are popular benchmarking software for storage devices. Their results can be compared with benchmarks conducted anywhere with similar versions of these software.

Features

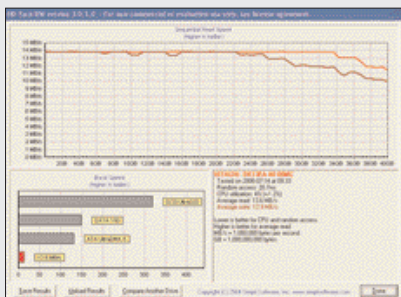
For features, we noted various aspects of the drives such as the capacity, dimensions, weight, bundled software, interface type, cables and accessories, and ruggedness.

Performance

Synthetic Test—HD Tach RW 3.0.1.0

The fresh hard drive to be tested was connected to the test machine and was not formatted; it was allowed to remain raw. HD Tach RW 3.0.1.0 was then run and used to evaluate the hard drive.

The scores we noted were as follows:



1. CPU Utilisation

This benchmark determines the CPU load of the storage device. Lower CPU utilisation is better. High CPU utilisation (above 15 per cent) generally indicates that a poor controller is being used, that DMA needs to be enabled, or that a driver update is required.

2. Random Access Time benchmark

This benchmark determines the random access speed of the device. Random access is the average time it takes to retrieve data from a randomly-located sector on the device. Lower random access speeds mean better application and database performance. Random access times can also give an insight into the efficiency of the interface a device is attached to.

3. Sequential Read and 4. Sequential Write

The maximum sustained speed (lowest speed on the device) is an important metric for real-time sequential access applications. The maximum speed may be important to a power user who would like to partition the drive into a high-speed partition and a low-speed partition. These were noted and logged.

Synthetic Test - SiSoft Sandra 2007 Engineer

A 20 GB FAT32 partition was created where possible, with the compression turned off, on the test hard drive. A second partition was created utilising the remaining capacity. SiSoft Sandra 2007 Engineer was launched and the File System benchmark module was run. The 20 GB partition we created was benchmarked using this module. The Sequential Read, Random

